

L-19

Ch-7.3

Simpson's Rule

We know

Trapezium Rule is for degree, $n = 1$.

If $n \uparrow$ then accuracy $\uparrow$.

So, $n = 2$ is more accurate.

Simpson's Rule is for degree, $n = 2$.

Previously, $n = 1$

$$I_1(f) = \int_a^b P_1(x) dx$$

$$\delta = \int_a^b \sum_{k=0}^1 \ell_k(x) dx$$

1 Derivation: Simpson's Rule

(degree is $n = 2$)

range = $[a, b]$

So,

$$x_0 = a$$

$$x_1 = m = \frac{a+b}{2}$$

$$x_2 = b$$

$$\begin{aligned} I_2(f) &= \sum_{k=0}^2 \delta_k f(x_k) \\ &= \delta_0 f(x_0) + \delta_1 f(x_1) + \delta_2 f(x_2) \end{aligned}$$

The Lagrange basis polynomials:

$$\ell_0(x) = \left(\frac{x - x_1}{x_0 - x_1} \right) \left(\frac{x - x_2}{x_0 - x_2} \right) = \frac{(x - m)(x - b)}{(a - m)(a - b)}$$

$$\ell_1(x) = \frac{x - x_0}{x_1 - x_0} \cdot \frac{x - x_2}{x_1 - x_2} = \frac{(x - a)(x - b)}{(m - a)(m - b)}$$

$$\ell_2(x) = \frac{x - x_0}{x_2 - x_0} \cdot \frac{x - x_1}{x_2 - x_1} = \frac{(x - a)(x - m)}{(b - a)(b - m)}$$

Computing the Omega values

$$\delta_0 = \int_a^b \ell_0(x) dx = \frac{1}{(a - m)(a - b)} \int_a^b (x - m)(x - b) dx$$

$$\int_a^b (x - m)(x - b) dx = \frac{2}{3} \left(\frac{b - a}{2} \right)^3 = \frac{2}{3} \cdot \frac{(b - a)^3}{8} = \frac{(b - a)^3}{12}$$

$$\delta_0 = \frac{2}{(b - a)^2} \cdot \frac{(b - a)^3}{12} = \frac{2(b - a)}{12} = \frac{b - a}{6}$$

$$\boxed{\therefore \delta_0 = \frac{1}{6}(b - a)}$$

$$\delta_1 = \int_a^b \ell_1(x) dx = \frac{1}{(m - a)(m - b)} \int_a^b (x - a)(x - b) dx$$

$$\therefore \delta_1 = \frac{2}{3}(b - a)$$

$$\delta_3 = \frac{1}{6}(b - a) \quad [\text{Similarly}]$$

Plugging the values

$$I_2(f) = \delta_0 f(x_0) + \delta_1 f(x_1) + \delta_2 f(x_2)$$

$$= \frac{1}{6}(b - a) f(x_0) + \frac{2(b - a)}{3} \cdot f(x_1) + \frac{1}{6}(b - a) \cdot f(x_2)$$

$$= \frac{b - a}{6} \left[f(a) + 4f\left(\frac{a + b}{2}\right) + f(b) \right]$$

$$\text{So, } I_2(f) = \frac{b - a}{6} \left[f(a) + 4f\left(\frac{a + b}{2}\right) + f(b) \right]$$

Example

$$f(x) = x^2 ; \quad [0, 2]$$

Here $a = 0, \quad b = 2$

$$\begin{aligned} I_2(f) &= \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] \\ &= \frac{2-0}{6} \left[0^2 + 4\left(\frac{0+2}{2}\right)^2 + 2^2 \right] \end{aligned}$$

$$\boxed{= \cancel{1}.\cancel{6}67}$$

$\textcolor{red}{2}$